

Deep Learning Labs

Sixteen standalone Google Colab activities built around evidence, reproducibility, and responsible practice

COURSE	Deep Learning AI-XXX-XXXXL
LAB CONTACT	3 hours per week 16 weeks 48 guided hours
ENVIRONMENT	Google Colab with PyTorch
NOTEBOOK MODE	Credential-free baseline; generated or embedded data
SUBMISSION	Executed notebook plus /content/artifacts/lab_XX/

Every lab uses the same learning cycle: predict, run, measure, explain, verify, save evidence, and reflect. FAST_MODE keeps the baseline suitable for Colab CPU; optional extensions are disabled by default.

How to use this manual

1. Open the matching notebook in Google Colab, select Runtime > Restart session, and run cells from the top.
2. Keep FAST_MODE = True for the first complete run. Use a GPU only when the notebook reports it is available.
3. Write the prediction before executing the principal experiment; do not revise it after seeing results.
4. Treat an assertion failure as diagnostic evidence. Read the nearby checkpoint before changing code.
5. Download both the executed notebook and the lab-specific artifact folder before the Colab runtime is recycled.
6. Complete optional extensions only after the baseline passes and preserve a controlled comparison.

Reproducibility and integrity standard

Control	Required practice	Evidence
Randomness	Retain the notebook seed and record device/framework versions.	Setup-cell output
Comparison	Change one principal factor at a time; reuse splits and budgets.	Paired metric table/figure
Validation	Run assertions and inspect shapes, ranges, and finite values.	Passing check cell
Interpretation	Link claims to a measured result and state a limitation.	Exit reflection
Responsible use	Label simulations; avoid unsupported claims about real groups or safety.	Model/result card

Suggested laboratory rubric

Criterion	Share	Full-credit evidence
Correct execution and checks	30%	Notebook runs top-to-bottom; baseline assertions pass.
Experimental design	20%	Prediction, controls, fair comparison, and reproducible settings.
Analysis and interpretation	25%	Metrics/figures are explained; claims match evidence.
Artifacts and communication	15%	Required files, labels, tables/cards, and readable presentation.
Limitations and reflection	10%	Failure mode, responsible-use caveat, and next experiment.
Assessment note The rubric is suggested. Apply departmental policy and disclose any local changes before students begin the assessed activity.		

Laboratory sequence

The sequence mirrors the updated course outline. Each notebook is standalone, but the habits of reproducibility, comparison, diagnostics, and limitation reporting accumulate across the semester.

Lab	Notebook focus	Lecture alignment	CLOs
01	PyTorch Tensors, Autograd, and Reproducible Training	Deep Learning Workflow and Representation Learning	CLO-1, CLO-2
02	Multilayer Perceptrons, Activations, and Losses	Multilayer Perceptrons, Activations, and Losses	CLO-1, CLO-2, CLO-3
03	Backpropagation and Gradient Diagnostics	Computational Graphs and Backpropagation	CLO-1, CLO-2, CLO-3
04	Optimization and Regularization	Optimization, Generalization, and Reliable Training	CLO-2, CLO-3, CLO-4
05	CNN Fundamentals and Feature Maps	Convolutional Neural Network Foundations	CLO-1, CLO-2, CLO-3
06	Residual Learning and Transfer Learning	Modern CNNs, Residual Learning, and Transfer	CLO-1, CLO-2, CLO-3, CLO-4
07	Object Detection and Semantic Segmentation	Object Detection and Semantic Segmentation	CLO-2, CLO-3, CLO-4
08	RNNs, LSTMs, and GRUs for Forecasting	Recurrent Networks and Gated Memory	CLO-1, CLO-2, CLO-3
09	Attention and a Tiny Transformer	Attention Mechanics and Transformer Blocks	CLO-1, CLO-2, CLO-3
10	Transformer Pretraining and Transfer	Pretraining and Transformer Model Families	CLO-1, CLO-2, CLO-3, CLO-4
11	Variational Autoencoders and Latent Spaces	Autoencoders and Variational Autoencoders	CLO-1, CLO-2, CLO-3
12	Generative Adversarial Networks	Generative Adversarial Networks	CLO-1, CLO-2, CLO-3, CLO-4
13	Toy Diffusion Models	Diffusion Models and Latent Generation	CLO-1, CLO-2, CLO-3
14	Graph Neural Networks from an Explicit Adjacency Matrix	Graph Neural Networks and Message Passing	CLO-1, CLO-2, CLO-3
15	Explainability, Fairness, and Responsible Model Cards	Explainability and Responsible Deep Learning	CLO-3, CLO-5
16	TorchScript Export, Latency, Drift, and Capstone Review	Deployment, Monitoring, and Project Defense	CLO-2, CLO-3, CLO-4, CLO-5

LAB 01 / THREE HOURS

PyTorch Tensors, Autograd, and Reproducible Training

LECTURE	Deep Learning Workflow and Representation Learning
CLO MAPPING	CLO-1, CLO-2

Learning objectives

- Manipulate tensor shapes, dtypes, and broadcasting safely.
- Verify an autograd result numerically.
- Implement and explain a reproducible PyTorch training loop.

Three-hour workflow

Time	Task
0-25 min	environment, tensor, dtype, and device checks
25-65 min	broadcasting and shape-debugging exercises
65-105 min	autograd and finite-difference verification
105-160 min	build, train, and plot linear regression
160-180 min	automated checks and reflection
	Predict before running Will autograd and a centered finite-difference estimate agree exactly? Predict the likely source and scale of any discrepancy.

Activities

Activity 1 | Tensors, broadcasting, and gradient verification

1. Create tensors with deliberate shapes and dtypes; annotate each axis.
2. Test valid and invalid broadcasting cases, then explain the resulting shape.
3. Compare autograd with an analytic or finite-difference gradient.

Activity 2 | A complete training loop

1. Build a full mini-batch training loop: forward, loss, zero_grad, backward, step.
2. Track loss and accuracy, save a learning-curve figure, and confirm repeatability with the fixed seed.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Completed tensor and gradient checks
- Loss curve and fitted-line artifact
- Saved state dictionary and explanation of zero_grad, backward, and step
- Executed notebook: Lab_01_PyTorch_Tensors_Autograd.ipynb

Book grounding

- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.

The notebook paraphrases concepts and provides original code; it does not reproduce book text.

Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 02 / THREE HOURS

Multilayer Perceptrons, Activations, and Losses

LECTURE	Multilayer Perceptrons, Activations, and Losses
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Build a multilayer perceptron with `nn.Module`.
- Relate activation and loss choices to task behavior.
- Compare models with controlled seeds, data, and training budgets.

Three-hour workflow

Time	Task
0-25 min	generate, split, and visualize data
25-65 min	activation/loss investigation
65-120 min	implement and train MLP variants
120-160 min	decision boundaries and metric comparison
160-180 min	checks and evidence-based recommendation
	Predict before running Which activation will reach the best test accuracy under the same budget, and why might the ranking change with a deeper network?

Activities

Activity 1 | Generate two moons and inspect activation functions

1. Generate a two-moons classification set and visualize the decision geometry.
2. Plot sigmoid, tanh, ReLU, and GELU with their useful and saturated regions.

Activity 2 | Controlled activation comparison

1. Train capacity-matched MLPs while changing only the activation.
2. Compare convergence, accuracy, and decision boundaries; justify the output-loss pairing.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Activation plot
- Executable MLP and decision-boundary plot
- Controlled comparison table and architecture recommendation
- Executed notebook: Lab_02_MLP_Activations_Losses.ipynb

Book grounding

- Goodfellow, Bengio, and Courville, Deep Learning, MIT Press, 2016.
- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.

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Extension and exit reflection

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LAB 03 / THREE HOURS

Backpropagation and Gradient Diagnostics

LECTURE	Computational Graphs and Backpropagation
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Verify chain-rule gradients against finite differences.
- Instrument layerwise gradient norms to identify vanishing or exploding gradients.
- Apply clipping and defensible debugging checks.

Three-hour workflow

Time	Task
0-35 min	chain-rule derivation and numerical check
35-75 min	hooks/gradient inspection
75-125 min	sigmoid versus ReLU deep-network diagnostic
125-160 min	clipped training and anomaly discussion
160-180 min	evidence capture and reflection
	Predict before running Predict which network, deep sigmoid or deep ReLU, will have the smaller gradient in its earliest layer. What would count as convincing evidence?

Activities

Activity 1 | Chain rule and finite-difference diagnostics

1. Evaluate a scalar computational graph by hand and with autograd.
2. Use finite differences to identify a deliberately incorrect derivative.

Activity 2 | Diagnose gradients in shallow and deep networks

1. Instrument layer-wise gradient norms in shallow and deep networks.
2. Relate activation and initialization choices to vanishing or exploding gradients.

Activity 3 | Gradient clipping and anomaly-safe training

1. Apply gradient clipping and anomaly checks to a stressed training run.
2. Report whether the intervention fixes the symptom or its underlying cause.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Finite-difference verification
- Layerwise gradient-norm visualization
- Clipped training run and diagnosis of one failure mode
- Executed notebook: Lab_03_Backprop_Gradient_Diagnostics.ipynb

Book grounding

- Goodfellow, Bengio, and Courville, Deep Learning, MIT Press, 2016.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.

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Extension and exit reflection

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Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 04 / THREE HOURS

Optimization and Regularization

LECTURE	Optimization, Generalization, and Reliable Training
CLO MAPPING	CLO-2, CLO-3, CLO-4

Learning objectives

- Compare SGD, momentum, RMSprop, and Adam under a matched budget.
- Apply normalization, dropout, weight decay, scheduling, and early stopping.
- Select and restore a model using validation rather than test results.

Three-hour workflow

Time	Task
0-30 min	data splits and leakage audit
30-90 min	optimizer benchmark
90-125 min	scheduler and normalization
125-155 min	dropout, weight decay, and early stopping
155-180 min	checkpoint, plots, and recommendation
	Predict before running Which optimizer will reduce training loss fastest, and will that necessarily identify the best regularized test model?

Activities

Activity 1 | A controlled optimization and regularization benchmark

1. Run a controlled optimizer/regularizer benchmark with identical splits and seeds.
2. Compare SGD, momentum, and Adam using validation evidence rather than final training loss alone.

Activity 2 | Learning curves, checkpointing, and augmentation reasoning

1. Plot learning curves, save the best checkpoint, and reason about weight decay, dropout, and augmentation.
2. Write an ablation conclusion that changes one factor at a time.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Matched optimizer comparison
- Regularization/early-stopping result
- Best checkpoint, metrics JSON, and learning-curve figure
- Executed notebook: Lab_04_Optimization_Regularization.ipynb

Book grounding

- Goodfellow, Bengio, and Courville, Deep Learning, MIT Press, 2016.
- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.

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Extension and exit reflection

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Exit reflection

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LAB 05 / THREE HOURS

CNN Fundamentals and Feature Maps

LECTURE	Convolutional Neural Network Foundations
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Calculate convolution/pooling shapes and receptive fields.
- Train a compact CNN on generated images.
- Compare CNN and MLP inductive biases using accuracy, parameters, and latency.

Three-hour workflow

Time	Task
0-30 min	image generation and kernel geometry
30-70 min	padding, stride, pooling, and receptive fields
70-130 min	CNN implementation and training
130-160 min	CNN/MLP comparison and feature maps
160-180 min	checks and interpretation
	Predict before running Will the CNN use fewer parameters than the MLP while matching its accuracy? Explain which architectural assumption supports your prediction.

Activities

Activity 1 | Generate images and inspect convolution geometry

1. Generate small images and calculate convolution output shapes before running code.
2. Measure how kernel, padding, and stride change spatial size and receptive field.

Activity 2 | Train a CNN and a parameter-matched MLP

1. Train a CNN and a parameter-matched MLP on the same split.
2. Compare accuracy, parameter count, and inductive bias.

Activity 3 | Feature-map visualization

1. Visualize early feature maps and connect responses to edges, textures, or regions in the input.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Shape and receptive-field calculations
- Trained CNN/MLP comparison JSON
- Feature-map visualization and architectural interpretation
- Executed notebook: Lab_05_CNN_Fundamentals_Feature_Maps.ipynb

Book grounding

- Goodfellow, Bengio, and Courville, Deep Learning, MIT Press, 2016.
- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.

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Extension and exit reflection

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Exit reflection

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LAB 06 / THREE HOURS

Residual Learning and Transfer Learning

LECTURE	Modern CNNs, Residual Learning, and Transfer
CLO MAPPING	CLO-1, CLO-2, CLO-3, CLO-4

Learning objectives

- Implement and shape-test a residual block.
- Pretrain on a self-generated source task and transfer representations to a small target task.
- Compare frozen, partially fine-tuned, and scratch strategies.

Three-hour workflow

Time	Task
0-25 min	source/target dataset audit
25-65 min	residual block and gradient pathway
65-110 min	source pretraining
110-155 min	frozen transfer and partial fine-tuning
155-180 min	scratch comparison, checks, and recommendation
	Predict before running Will transferred features help more than scratch training when only 80 target examples are available? State what outcome would falsify your prediction.

Activities

Activity 1 | Synthetic source and target domains

1. Create related source and target image domains and quantify their shift.

Activity 2 | Residual block and source pretraining

1. Implement a residual block, verify the skip-path shapes, and pretrain on the source task.

Activity 3 | Frozen transfer, partial fine-tuning, and scratch baseline

1. Compare frozen transfer, partial fine-tuning, and training from scratch on the target set.
2. Use the same target split and report quality versus trainable parameters.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Residual-block unit test
- Source checkpoint and fine-tuned target checkpoint
- Transfer-versus-scratch plot with accuracy/parameter comparison
- Executed notebook: Lab_06_Residual_and_Transfer_Learning.ipynb

Book grounding

- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

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Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 07 / THREE HOURS

Object Detection and Semantic Segmentation

LECTURE	Object Detection and Semantic Segmentation
CLO MAPPING	CLO-2, CLO-3, CLO-4

Learning objectives

- Represent class, bounding-box, and mask targets correctly.
- Train a compact multi-task vision model.
- Evaluate detection and segmentation with class accuracy, box IoU, mask IoU, and Dice.

Three-hour workflow

Time	Task
0-30 min	target formats and synthetic data
30-65 min	encoder, heads, and multi-task losses
65-125 min	train and debug
125-160 min	IoU/Dice evaluation and overlays
160-180 min	error analysis and checks
	Predict before running Which task head-classification, box regression, or pixel segmentation-will learn fastest on this dataset, and what metric will reveal it?

Activities

Activity 1 | Generate images with class, box, and mask targets

1. Generate images with aligned class, bounding-box, and segmentation-mask targets.
2. Visualize target overlays and verify coordinate conventions.

Activity 2 | Train a multi-task detector/segmenter

1. Train the multi-task model and track classification, box, and mask losses separately.

Activity 3 | Dice, mask IoU, box IoU, and overlays

1. Compute box IoU, mask IoU, and Dice; inspect qualitative overlays and categorize errors.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Verified target tensors
- Executable multi-task model
- Metrics JSON and qualitative overlay grid
- Short comparison of classification, detection, and segmentation outputs
- Executed notebook: Lab_07_Detection_Segmentation.ipynb

Book grounding

- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

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Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 08 / THREE HOURS

RNNs, LSTMs, and GRUs for Forecasting

LECTURE	Recurrent Networks and Gated Memory
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Create chronological sequence windows without leakage.
- Implement and compare vanilla RNN, LSTM, and GRU models.
- Relate state structure and gradient behavior to forecasting performance.

Three-hour workflow

Time	Task
0-30 min	time-series split and windowing
30-65 min	hidden/cell state shape exercises
65-125 min	train three recurrent models
125-160 min	forecasts, gradients, cost comparison
160-180 min	checks and architecture recommendation
	Predict before running Which gated recurrent model will achieve the lowest RMSE, and does its extra parameter count justify the difference?

Activities

Activity 1 | Chronological windows without leakage

1. Create chronological windows with no future leakage and verify indices explicitly.

Activity 2 | Capacity-matched RNN, LSTM, and GRU

1. Train capacity-matched RNN, LSTM, and GRU models with the same windows and stopping rule.

Activity 3 | Forecast comparison

1. Plot forecasts, compare error and parameter count, and identify where recurrent memory helps or fails.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Leakage-safe windowing assertions
- RNN/LSTM/GRU metric table
- Forecast/learning-curve artifact and recommendation
- Executed notebook: Lab_08_RNN_LSTM_GRU.ipynb

Book grounding

- Goodfellow, Bengio, and Courville, Deep Learning, MIT Press, 2016.
- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.

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Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 09 / THREE HOURS

Attention and a Tiny Transformer

LECTURE	Attention Mechanics and Transformer Blocks
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Implement scaled dot-product attention and causal masking.
- Use positional representations and multi-head self-attention in a tiny Transformer.
- Interpret attention weights cautiously and distinguish common Transformer objectives.

Three-hour workflow

Time	Task
0-35 min	attention calculation and unit tests
35-65 min	masking and positional encoding
65-120 min	Tiny Transformer implementation/training
120-155 min	attention visualization and errors
155-180 min	BERT/GPT/T5 objective comparison and checks
	Predict before running Will the first query position learn to attend strongly to the last key position? Explain why a visible weight would still not prove causal importance.

Activities

Activity 1 | Scaled dot-product attention, masks, and positional information

1. Implement scaled dot-product attention and verify tensor shapes.
2. Test padding/causal masks and positional information with small inspectable examples.

Activity 2 | Tiny Transformer for a sequence-relation task

1. Train a tiny Transformer on a synthetic sequence-relation task and compare against a simple baseline.

Activity 3 | Inspect a head, then distinguish BERT-, GPT-, and T5-style objectives

1. Inspect one attention head without treating attention as a causal explanation.
2. Classify example objectives as BERT-, GPT-, or T5-style.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Attention/mask unit tests
- Trained tiny Transformer checkpoint
- Loss/attention figure
- BERT-, GPT-, and T5-style objective comparison
- Executed notebook: Lab_09_Attention_Tiny_Transformer.ipynb

Book grounding

- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

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Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 10 / THREE HOURS

Transformer Pretraining and Transfer

LECTURE	Pretraining and Transformer Model Families
CLO MAPPING	CLO-1, CLO-2, CLO-3, CLO-4

Learning objectives

- Construct a masked-token self-supervised objective.
- Pretrain a toy Transformer encoder without external weights or data.
- Transfer its frozen representation to a small labeled task and compare with scratch.

Three-hour workflow

Time	Task
0-25 min	corpus/objective construction
25-80 min	masked-token encoder and pretraining
80-115 min	representation checks
115-155 min	frozen transfer and scratch baseline
155-180 min	comparison, leakage audit, and reflection
	Predict before running Predict whether masked-token pretraining will improve data efficiency on the direction task, and identify a possible mismatch between the pretraining and transfer objectives.

Activities

Activity 1 | Create a toy sequence corpus and masked-token pretraining task

1. Generate a toy sequence corpus, mask tokens without leaking targets, and define the pretraining objective.

Activity 2 | Pretrain an encoder to reconstruct masked tokens

1. Pretrain a compact encoder and track masked-token reconstruction performance.

Activity 3 | Transfer to direction classification and compare with scratch

1. Transfer the encoder to direction classification and compare with a scratch model under the same labeled-data budget.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Masked-pretraining dataset and checks
- Pretraining/transfer curves
- Transferred checkpoint
- Transfer-versus-scratch result with limitations
- Executed notebook: Lab_10_Transformer_Pretraining_Transfer.ipynb

Book grounding

- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

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Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 11 / THREE HOURS

Variational Autoencoders and Latent Spaces

LECTURE	Autoencoders and Variational Autoencoders
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Implement VAE reparameterization and ELBO components.
- Train a compact VAE and inspect reconstructions.
- Analyze latent organization and the reconstruction-regularization trade-off.

Three-hour workflow

Time	Task
0-30 min	autoencoder baseline and probabilistic notation
30-70 min	encoder, reparameterization, decoder
70-125 min	ELBO implementation and training
125-160 min	reconstruction/latent analysis
160-180 min	checks and comparison
	Predict before running As KL weight increases, predict what happens to reconstruction sharpness and latent organization. Which plot would support each claim?

Activities

Activity 1 | Build a small image distribution

1. Generate a compact image distribution and inspect its class/variation structure.

Activity 2 | Reparameterization and ELBO training

1. Implement reparameterization and train with reconstruction plus KL terms.
2. Plot both ELBO components and diagnose imbalance or posterior collapse.

Activity 3 | Reconstructions, latent space, and interpolation

1. Compare inputs and reconstructions, visualize latent codes, and generate an interpolation path.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- VAE implementation with reconstruction and KL terms
- Reconstruction and latent artifacts
- Saved VAE state and trade-off explanation
- Executed notebook: Lab_11_VAE_Latent_Spaces.ipynb

Book grounding

- Goodfellow, Bengio, and Courville, Deep Learning, MIT Press, 2016.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

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Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 12 / THREE HOURS

Generative Adversarial Networks

LECTURE	Generative Adversarial Networks
CLO MAPPING	CLO-1, CLO-2, CLO-3, CLO-4

Learning objectives

- Implement alternating discriminator and generator updates.
- Diagnose mode coverage and unstable adversarial losses.
- Generate low-resolution image samples with a CPU-friendly GAN.

Three-hour workflow

Time	Task
0-30 min	minimax objective and alternating updates
30-90 min	train 2-D multimodal GAN
90-125 min	mode-collapse diagnostics
125-160 min	image-GAN smoke test
160-180 min	stabilization analysis and checks
	Predict before running Do you expect discriminator and generator losses to move monotonically? Predict a mode-coverage result and explain why loss alone may mislead.

Activities

Activity 1 | Adversarial learning on an eight-mode distribution

1. Train a GAN on an eight-mode 2-D distribution with alternating discriminator and generator updates.
2. Track losses and mode coverage rather than trusting either loss alone.

Activity 2 | A tiny image GAN smoke test

1. Run the tiny image-GAN smoke test and verify generated tensor ranges and shapes.

Activity 3 | Mode coverage and generated-sample diagnostics

1. Visualize samples, quantify covered modes, and document signs of imbalance or collapse.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Correctly detached discriminator update
- Mode-coverage/loss artifact
- Generated image grid and saved generator
- Brief diagnosis of instability or collapse
- Executed notebook: Lab_12_Generative_Adversarial_Networks.ipynb

Book grounding

- Goodfellow, Bengio, and Courville, Deep Learning, MIT Press, 2016.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

The notebook paraphrases concepts and provides original code; it does not reproduce book text.

Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 13 / THREE HOURS

Toy Diffusion Models

LECTURE	Diffusion Models and Latent Generation
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Implement a forward Gaussian-noising schedule.
- Train a timestep-conditioned network to predict noise.
- Run and visualize an approximate reverse-diffusion sampler.

Three-hour workflow

Time	Task
0-30 min	variance schedule and forward process
30-65 min	timestep conditioning and loss
65-120 min	train noise predictor
120-160 min	reverse sampler and trajectories
160-180 min	compare assumptions, checks, and reflection
	Predict before running At which forward timestep will the four clusters become visually indistinguishable? Predict whether low noise-prediction MSE guarantees good samples.

Activities

Activity 1 | Forward diffusion on a multimodal distribution

1. Apply the forward noising process at selected timesteps and visualize the loss of structure.

Activity 2 | Train a timestep-conditioned noise predictor

1. Train a timestep-conditioned network to predict added noise and compare error across timesteps.

Activity 3 | Reverse sampling and denoising trajectory

1. Run reverse sampling, save a denoising trajectory, and compare generated versus target distributions.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Forward-process snapshots
- Noise-prediction learning evidence
- Reverse-sampling trajectory and saved predictor
- Brief comparison with VAE and GAN sampling
- Executed notebook: Lab_13_Toy_Diffusion.ipynb

Book grounding

- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

The notebook paraphrases concepts and provides original code; it does not reproduce book text.

Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 14 / THREE HOURS

Graph Neural Networks from an Explicit Adjacency Matrix

LECTURE	Graph Neural Networks and Message Passing
CLO MAPPING	CLO-1, CLO-2, CLO-3

Learning objectives

- Represent an undirected graph with explicit adjacency and feature matrices.
- Implement symmetric normalization and two GCN layers using native PyTorch.
- Compare graph-aware message passing with an edge-ignorant baseline.

Three-hour workflow

Time	Task
0-30 min	graph representation and node/graph task distinction
30-70 min	self-loops and normalized adjacency
70-125 min	implement/train GCN
125-155 min	MLP baseline and visualization
155-180 min	GraphSAGE/GAT comparison and checks
	Predict before running Will a GCN classify unlabeled nodes better than an MLP that ignores edges? Identify how graph homophily affects that prediction.

Activities

Activity 1 | Explicit graph tensors and normalized message passing

1. Represent a small graph with features, labels, and an explicit adjacency matrix.
2. Add self-loops and verify normalized message passing numerically.

Activity 2 | Native-PyTorch GCN and edge-ignorant MLP

1. Implement a two-layer GCN in native PyTorch and compare it with an edge-ignorant MLP.

Activity 3 | Visualize message-passing predictions

1. Plot node predictions on the graph and analyze helpful propagation, oversmoothing, and split limitations.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Explicit adjacency and normalization checks
- Native-PyTorch GCN implementation - no torch-geometric
- GCN/MLP comparison and graph visualization
- Executed notebook: Lab_14_Graph_Neural_Networks.ipynb

Book grounding

- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.
- Prince, Understanding Deep Learning, MIT Press, 2023.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.

The notebook paraphrases concepts and provides original code; it does not reproduce book text.

Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 15 / THREE HOURS

Explainability, Fairness, and Responsible Model Cards

LECTURE	Explainability and Responsible Deep Learning
CLO MAPPING	CLO-3, CLO-5

Learning objectives

- Generate input-gradient and Grad-CAM explanations with native PyTorch.
- Audit demographic-parity, true-positive-rate, and false-positive-rate gaps on a labeled simulation.
- Evaluate a mitigation trade-off and document limitations in a model card.

Three-hour workflow

Time	Task
0-35 min	train vision model and saliency
35-70 min	Grad-CAM and occlusion faithfulness
70-110 min	simulated fairness dataset/model
110-145 min	subgroup metrics and mitigation
145-180 min	model card, ethical limitations, and checks
	Predict before running Predict whether deleting the most salient pixels will reduce confidence more than deleting random pixels, and predict the fairness-gap direction created by the simulated label rule.

Activities

Activity 1 | Native saliency and Grad-CAM on generated images

1. Generate input-gradient saliency and Grad-CAM for the same prediction.
2. Use occlusion as a limited faithfulness check and state why the map is not causal proof.

Activity 2 | Fairness audit on a pedagogical simulation

1. Audit selection rate, TPR, and FPR by group on the explicitly labeled pedagogical simulation.
2. Evaluate a threshold mitigation and its accuracy trade-off without generalizing to real people.

Activity 3 | Evidence figures and model card

1. Save the explanation figure and fairness report; write a model card with intended use, data, evidence, and limitations.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Saliency/Grad-CAM evidence and occlusion result
- Fairness audit JSON with before/after trade-off
- Model card explicitly identifying simulated data and non-generalizability
- Executed notebook: Lab_15_XAI_Fairness_Model_Card.ipynb

Book grounding

- Prince, Understanding Deep Learning, MIT Press, 2023.
- Thampi, Interpretable AI, Manning, 2022.
- Barocas, Hardt, and Narayanan, Fairness and Machine Learning, MIT Press, 2023.

The notebook paraphrases concepts and provides original code; it does not reproduce book text.

Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

LAB 16 / THREE HOURS

TorchScript Export, Latency, Drift, and Capstone Review

LECTURE	Deployment, Monitoring, and Project Defense
CLO MAPPING	CLO-2, CLO-3, CLO-4, CLO-5

Learning objectives

- Run a reproducible paired model comparison with quality and cost metrics.
- Export and verify a TorchScript artifact without a live server.
- Measure latency and simulated drift, then package a responsible result card.

Three-hour workflow

Time	Task
0-35 min	clean-run and paired experiment
35-75 min	quality, parameters, and latency comparison
75-110 min	TorchScript export/equivalence
110-145 min	drift/monitoring analysis
145-180 min	result card, demonstration, peer review, and course reflection
	Predict before running Assess whether the nonlinear MLP will justify its extra parameters and latency, predict how the simulated feature shift will alter confidence, and define an alert threshold.

Activities

Activity 1 | Reproducible paired capstone experiment

1. Run a paired linear-versus-MLP experiment with fixed data, seeds, quality metrics, parameter count, and latency.

Activity 2 | TorchScript export and equivalence

1. Export the selected model to TorchScript, reload it, and verify numerical equivalence.

Activity 3 | Latency, drift, robustness, and capstone result card

1. Simulate feature drift, define a monitoring signal, and package the decision in a result card.
2. Demonstrate the artifact and defend the choice using measured evidence and stated limitations.

Verification checkpoint

- Restart the runtime and run all baseline cells in order.
- Confirm every automated assertion passes and no output contains NaN or Inf.
- Check that the expected files exist in the lab artifact directory.
- Record at least one measured value and explain whether it supports the initial prediction.

Submission evidence

- Restart-and-run-all paired experiment
- TorchScript model and equivalence evidence
- Latency/drift visualization
- Result card with explicit limitations and capstone recommendation
- Executed notebook: Lab_16_Export_Latency_Drift_Capstone.ipynb

Book grounding

- Huyen, Designing Machine Learning Systems, O'Reilly Media, 2022.
- Bishop and Bishop, Deep Learning: Foundations and Concepts, Springer, 2024.
- Zhang, Lipton, Li, and Smola, Dive into Deep Learning, Cambridge University Press, 2024.

The notebook paraphrases concepts and provides original code; it does not reproduce book text.

Extension and exit reflection

- Complete the disabled extension only after the CPU baseline passes. State which variable you changed and preserve the baseline for comparison.

Exit reflection

In 4-6 sentences, state whether the prediction was supported, identify the strongest measured evidence, name one limitation or failure mode, and propose the next controlled experiment.

End-of-semester submission checklist

- ☐ All 16 notebooks are present, named consistently, and retain their setup/seed cells.
- ☐ Every baseline notebook runs from a clean runtime and completes its automated checks.
- ☐ Artifact folders contain the required figures, metric files, model artifacts, and cards.
- ☐ The capstone comparison uses a defensible metric, reports latency/cost, and records drift evidence.
- ☐ Interpretability and fairness evidence is labeled accurately, with simulations clearly identified.
- ☐ The final reflection distinguishes measured findings from assumptions and unsupported generalization.

Book code key

- B1** Prince, S. J. D. Understanding Deep Learning. 1st ed. The MIT Press, 2023.
- B2** Goodfellow, I., Bengio, Y., and Courville, A. Deep Learning. 1st ed. The MIT Press, 2016.
- B3** Zhang, A., Lipton, Z. C., Li, M., and Smola, A. J. Dive into Deep Learning. 1st ed. Cambridge University Press, 2024.
- B4** Bishop, C. M., and Bishop, H. Deep Learning: Foundations and Concepts. 1st ed. Springer, 2024.
- B5** Foster, D. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play. 2nd ed. O'Reilly Media, 2023.
- B6** Thampi, A. Interpretable AI: Building Explainable Machine Learning Systems. 1st ed. Manning, 2022.
- B7** Huyen, C. Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications. 1st ed. O'Reilly Media, 2022.
- B8** Barocas, S., Hardt, M., and Narayanan, A. Fairness and Machine Learning: Limitations and Opportunities. 1st ed. The MIT Press, 2023.